STA 325 Case Study Report

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Introduction

This case study explores how three factors affect the clustering of inertial particles in turbulent flows such as clouds. The data come from computer simulations: in each run, our collaborators simulated particles moving in a turbulent flow, identified clusters of particles, and measured the volume of each cluster. The dataset has 89 simulation runs, each summarizing the distribution of particle cluster volumes at one (Re, Fr, St) setting. The three parameters take the following values:

- **Reynolds number** Re: 90, 224, 398. This measures how strong the turbulence is.
- **Froude number** Fr: 0.052, 0.3, ∞. This measures the strength of gravity. Smaller Fr means stronger gravity, and $Fr = \infty$ means no gravity.
- **Stokes number** St: 14 values between 0.05 and 3. This measures particle inertia. Particles with small St follow the flow closely, while particles with large St are heavier and respond more slowly.

Our key research objectives are:

1. **Prediction:** predict the first 4 moments of the cluster-volume distribution at new (Re, Fr, St) settings as a fast substitute for running a new simulation. Full computer simulations are expensive and slow: a full run for a single (Re, Fr, St) setting can take days or longer, while a fitted statistical model returns a prediction in seconds. A good predictive model lets our collaborators explore new parameter settings almost instantly instead of waiting for days.
2. **Inference:** Explaining how each parameter affects the response

Our final model is a separate generalized additive model for each response. Here Re_{224}, Re_{398}, $Fr_{0.3}$, and Fr_{∞} are 0/1 indicators for each level (baseline $Re = 90$, $Fr = 0.052$), and f is one smooth function shared across the observed (Re, Fr) settings.

$$\log \text{response} = \beta_0 + \alpha_{Re} + \gamma_{Fr} + \delta_{Re \times Fr} + f(\log St) + \varepsilon, \qquad \varepsilon \sim N(0, \sigma^2).$$

A separate model is fit for each of the 4 response variables, so each one has its own coefficients and smooth. There is no $Re_{398}Fr_{0.3}$ term because the dataset has no observations at $Re = 398$, $Fr = 0.3$.

This model addresses both objectives. For prediction, it achieves a LOOCV MSE of 0.087 when averaged across the 4 response variables. For inference, the categorical terms describe differences among Re and Fr settings, while the smooth allows the effect of St to vary across its range.

Methodology

Responses and predictors

Each observation in the dataset reports the first four raw moments of the response distribution, which we converted to central moments with `raw2central`. From these we computed the mean, standard deviation, skewness, and kurtosis of each distribution, which are the 4 desired response variables.

Transformations

The 4 responses differ in range and in skewness, so we checked each separately, before and after a log transform (Figure 3 in the appendix). The mean and standard deviation span 776-fold and 3,212-fold ranges and are strongly right-skewed, with skewness 0.97 and 2.44. The log reduces both, to 0.22 and 0.99, although the standard deviation stays moderately right-skewed. Skewness and kurtosis behave differently. Skewness spans only a 29-fold range and is nearly symmetric (-0.03), and kurtosis spans a 878-fold range but is only mildly right-skewed (0.40). The log over-corrects both into left skew (-0.78 and -0.79). A single log transform therefore might not suit all four responses equally well.

To investigate this, we fit separate Box-Cox transforms to each response. The resulting $\hat{\lambda}$ ranges from -0.11 to -0.02, and every 95% interval contains 0 (Table 5 and Figure 4), which corresponds to log transformations, so the data supports the use of log transformations on all 4 responses.

We also log-transformed St because it is right-skewed (skewness of 1.43), spans a wide range, and generally yielded better empirical models when transformed.

Modeling

We compared the cubic polynomial to natural cubic splines and GAMs. Cubic polynomials can behave poorly near the boundary of the design, whereas a natural spline is constrained to be linear beyond its boundary knots and a GAM controls curvature with a smoothing penalty. All nonlinear candidates use a common St curve plus a $Re \times Fr$ interaction. We deliberately did not retain the earlier group-specific GAM, which estimated a separate curve for each (Re, Fr) group and was too flexible for only 14 observed St values.

All models predict the log of each response. The baseline model has no interaction terms. Interaction Model 1 adds interactions of $\log St$ with both Re and Fr, as well as $Re \times Fr$. Interaction Model 2 adds only $Re \times Fr$. The cubic and natural-spline candidates add increasingly flexible common St effects to Interaction Model 2. We tuned natural-spline degrees of freedom (3, 4, and 5) and the GAM basis dimension ($k = 4, 5, 6$) by LOOCV; REML selects the GAM's smoothing penalty within each training fold.

Averaged across the 4 response variables, the LOOCV MSE was 0.609 for the baseline model, 0.122 for the $Re \times Fr$ model, 0.088 for the cubic, and 0.087 for the selected GAM. The GAM with $k = 5$ had the lowest average LOOCV MSE and avoided the cubic's boundary behavior, so we selected it as our final model. The table reports the full tuning comparison for each response.

For the GAM, the parameter count is its effective degrees of freedom; it is reported for the mean model as a compact reference because REML can select slightly different smoothness for the other responses.

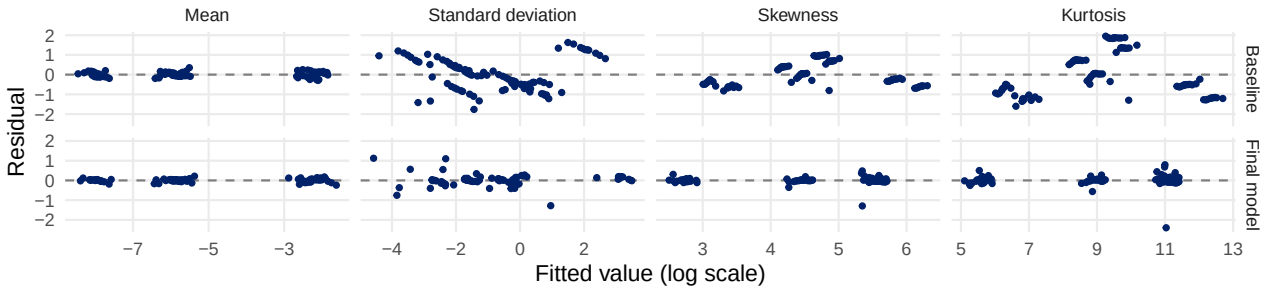
Table 1: Comparison of LOOCV MSE accross different models

Model	Parameters	Mean	Standard deviation	Skewness	Kurtosis
Baseline (additive)	6.0	0.0192	0.7338	0.3561	1.3252
Interaction Model 1	13.0	0.0104	0.3290	0.0715	0.2119
Interaction Model 2	9.0	0.0114	0.2565	0.0554	0.1656
Cubic Model	11.0	0.0093	0.1377	0.0480	0.1567
Natural spline (3 df)	11.0	0.0095	0.1565	0.0479	0.1542
Natural spline (4 df)	12.0	0.0090	0.1447	0.0484	0.1569
Natural spline (5 df)	13.0	0.0092	0.1413	0.0487	0.1589
Common GAM (k = 4)	10.9	0.0090	0.1451	**0.0478**	**0.1541**
Common GAM (k = 5)	11.7	**0.0089**	**0.1356**	0.0484	0.1559
Common GAM (k = 6)	12.1	0.0091	0.1359	0.0487	0.1567

Re and Fr each have only 3 levels, so we treat them as categorical.

Diagnostics

The selected GAM fits all 4 response variables well, with R^2 between 0.972 and 0.999. The plots below compare the residuals of the baseline model (top) and the selected GAM (bottom).



The baseline residuals show clear patterns, especially for skewness and kurtosis, where whole groups of observations sit above or below 0. This means the effect of Re depends on Fr. The selected GAM accounts for this interaction and curvature in St; its residuals are centered around 0, although some variation in spread remains, particularly for standard deviation.

The residual distributions still depart from normality, especially in their tails. The penalized smooth allows the fitted relationship to bend across the observed St range, but observations at $St = 0.05$ remain worth monitoring because this is the boundary of the design. We kept these observations because there is no sign they are errors.

The dataset was collected from computer simulation runs, but observations with the same Re and Fr likely come from the same simulation, so they may not be fully independent. This could make our uncertainty estimates slightly too small.

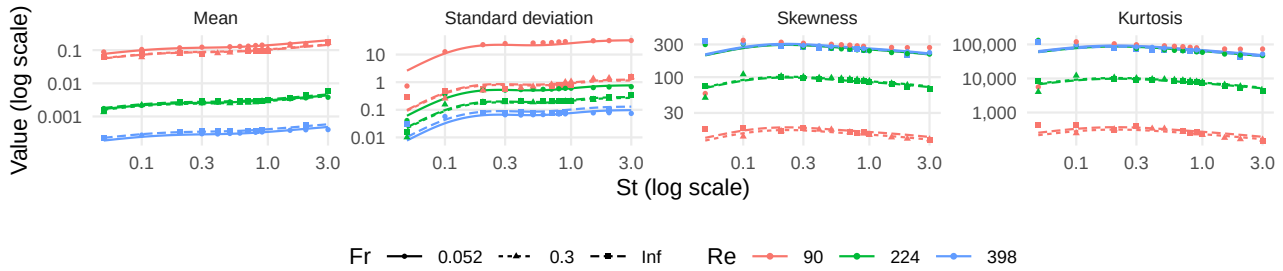


Figure 1: Fitted values from the final model (lines) and observed data (points). Both axes are on a log scale.

Results

Predictions on the hold-out set

We predicted the four statistics at the 23 hold-out settings, with 95% prediction intervals computed on the log scale and then exponentiated, so point predictions estimate medians. All values are provided in `predictions.csv`. Every hold-out setting uses an (Re, Fr) combination and St value within the training range, so no prediction requires extrapolation. The two settings with $St = 0.05$ lie at the boundary of the observed range and should be interpreted with that limitation in mind.

The precision of the predictions varies across the four statistics. Across the hold-out settings, the upper bounds are about 1.4 times the lower bounds for the mean, 3.7 times for standard deviation, 2.2 times for skewness, and 4.1 times for kurtosis. The mean is predicted most precisely, while the standard deviation and kurtosis intervals are widest.

Effects of each parameter

Because the model is fit on the log scale, exponentiating a contrast gives the multiplicative change in a statistic. Since Re and Fr interact, the gravity contrast at a given Re combines the Fr coefficient with the corresponding interaction coefficient. The smooth term means the effect of St varies across its range rather than remaining a constant multiplier.

Turbulence intensity (Re) controls cluster size. The GAM's fitted contrasts show that increasing Re from 90 to 398 markedly shrinks mean cluster volume and standard deviation, with the reduction in standard deviation weaker under no gravity. Thus, Re has a large association with cluster scale. Under strong gravity, its estimated effects on skewness and kurtosis are comparatively small in this model.

Strong gravity creates heavy tails, but only when turbulence is weak. At $Re = 90$, the fitted gravity contrasts show much higher skewness and kurtosis under strong gravity ($Fr = 0.052$) than under no gravity ($Fr = \infty$). Strong gravity is therefore associated with a distribution having many small clusters and occasional very large ones. This effect weakens as Re increases. One physical interpretation is that as turbulence strengthens, turbulent velocity fluctuations overwhelm gravitational settling, so the flow rather than gravity determines how particles cluster.

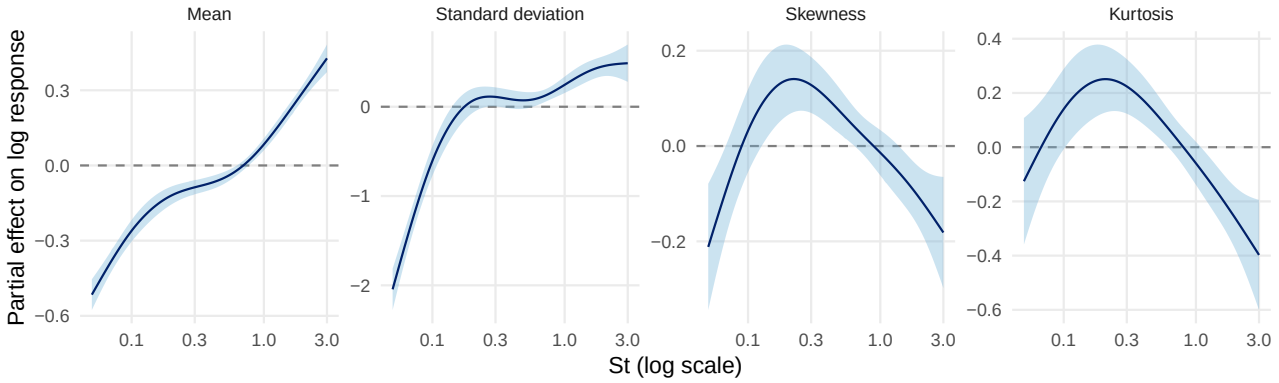


Figure 2: Estimated partial effect of particle inertia on the log of each response. Shading is an approximate 95% confidence band; zero is the average smooth effect.

Moderate gravity behaves like no gravity. In the GAM, the fitted effects for $Fr = 0.3$ and $Fr = \infty$ are similar across the four responses. Within the range studied, gravity is most strongly associated with clustering outcomes when it is strong.

Particle inertia (St) has a curved association with the responses. The selected smooths (Figure Figure 2) allow the relationship between St and each statistic to change in slope across the observed range. A single percentage change for doubling St would therefore be misleading; the confidence bands show where each response-specific curve is most and least precisely estimated. The model uses a common curve across (Re, Fr) settings, while allowing the levels of the curves to differ by those settings.

Conclusion

Each parameter shapes the cluster-size distribution in a different way. Turbulence intensity is strongly associated with cluster scale, while strong gravity is associated with heavier tails when turbulence is weak. The GAM also indicates that the association with particle inertia changes across the observed St range. Because gravity's influence varies with turbulence strength, conclusions drawn at one turbulence level should not be assumed to hold at another. The model predicts mean cluster volume most precisely; uncertainty is wider for the other statistics.

Limitations and future work

The model can only predict at the 3 simulated levels of Re and Fr. Because it treats Re and Fr as categorical, it cannot interpolate intermediate values between levels or extrapolate to more extreme values than simulated. Mapping an unseen input level to the nearest simulated level would let the model perform prediction, but the offset would be completely discarded, and per-unit inference would still be impossible. The model also has no runs at $Re = 398$, $Fr = 0.3$, so interaction effects at that combination cannot be modeled.

Future simulations should be spread off the grid. The current runs form a 3 by 3 grid in (Re, Fr) with one combination missing. A more random non-grid structure to selecting input values would be beneficial, allowing Fr and Re to be treated as continuous, which they are. This change

will improve interpolation and extrapolation. To address $Fr = \infty$ on a continuous scale, a $1/Fr$ transformation could be introduced.

Appendix: Exploratory Data Analysis

Data Processing

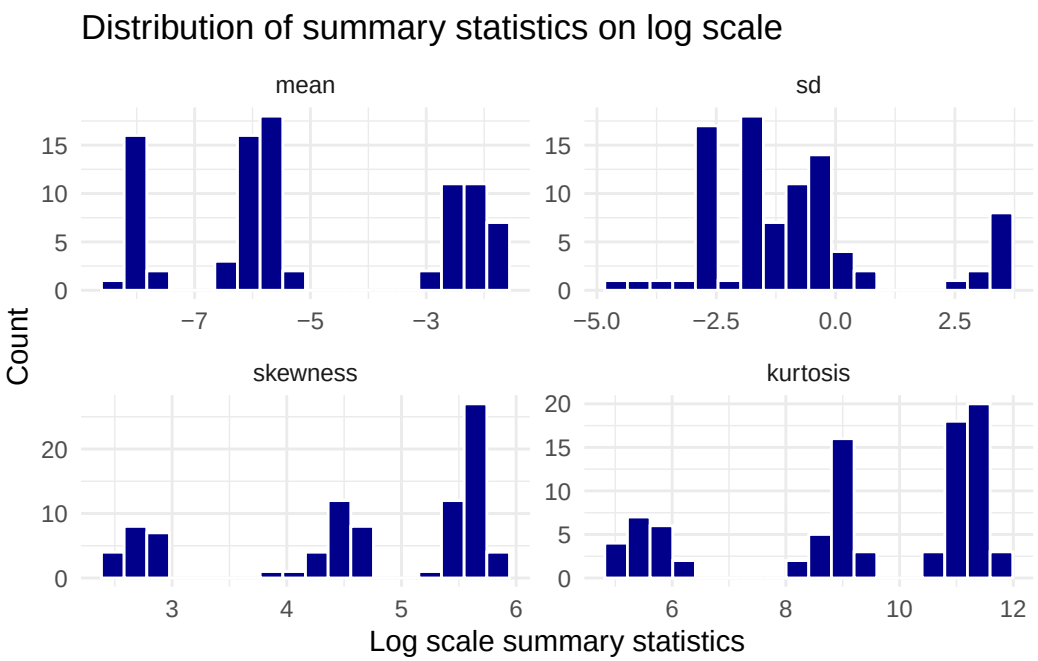
We see that the existing data contains Re and Fr values that are collected at distinct levels with low cardinality, instead of continuous values. It includes a valid infinity value for one of the predictors that is not a result of missing data. We compute standard deviation, skewness, and kurtosis from raw moments.

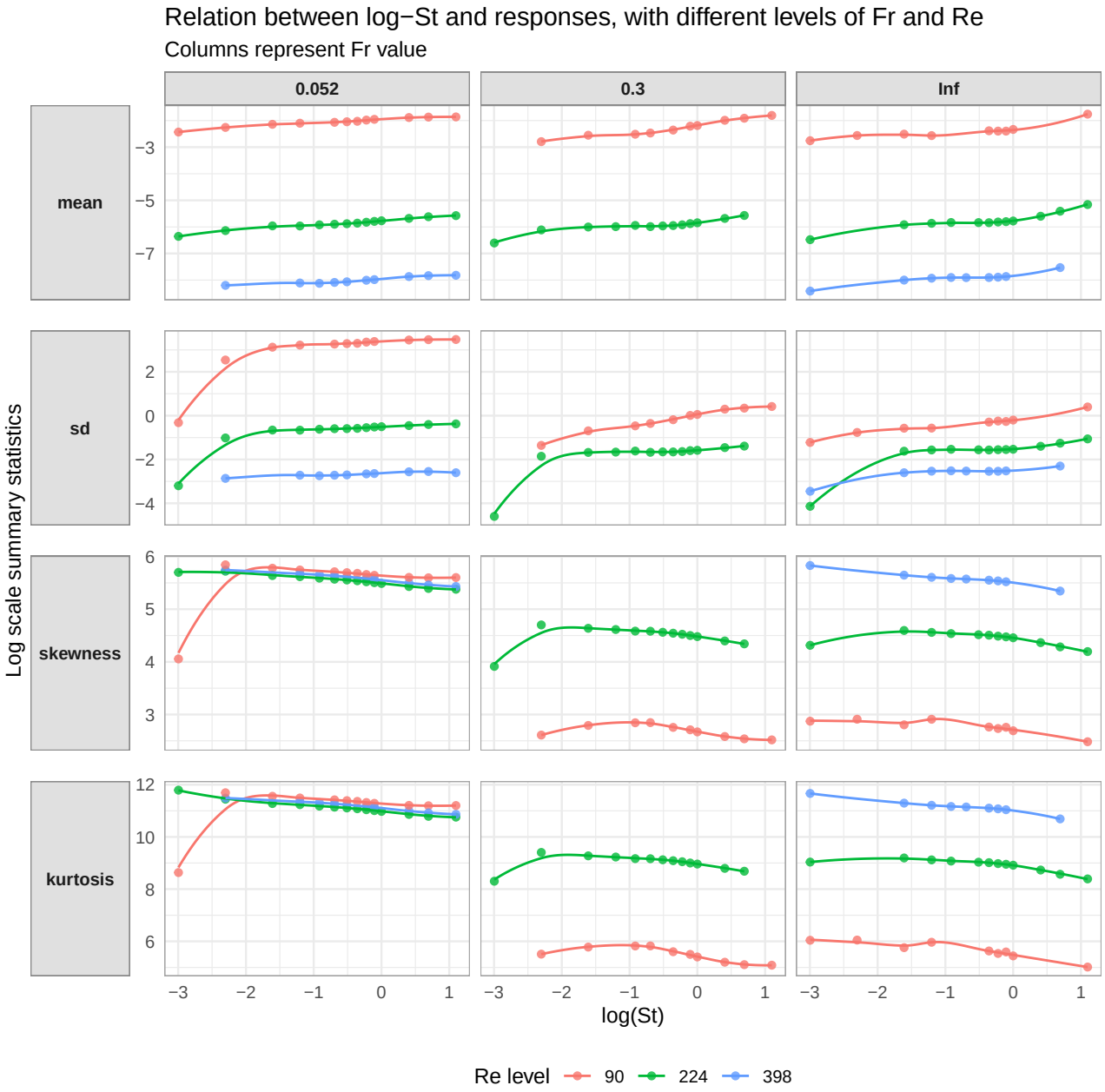
predictor	level	n
Re	224	39
Re	398	19
Re	90	31
Fr	0.052	36
Fr	0.3	23
Fr	Inf	30

response	min	q1	median	mean	q3	max	sd
mean	0.00	0.00	0.00	0.04	0.09	0.17	0.06
sd	0.01	0.16	0.28	3.65	0.73	32.32	8.95
skewness	11.97	72.55	110.12	162.81	269.54	344.91	113.10
kurtosis	150.53	5622.34	12158.72	39749.56	72732.45	132136.73	38100.12

Exploratory Data Analysis

Re	Fr_condition	n	St_min	St_max	St_levels
90	0.052	12	0.05	3	12
90	0.3	10	0.10	3	10
90	Inf	9	0.05	3	9
224	0.052	14	0.05	3	14
224	0.3	13	0.05	2	13
224	Inf	12	0.05	3	12
398	0.052	10	0.10	3	10
398	Inf	9	0.05	2	9


Distribution of summary statistics on log scale
mean
sd
skewness
kurtosis
Count
Log scale summary statistics



Response transformations

Table 5: Estimated Box-Cox λ for each response, with 95% profile-likelihood intervals. $\lambda = 0$ is the log transform.

Response	Estimate	95% interval
Mean	-0.02	[-0.05, 0.01]
Standard deviation	-0.03	[-0.07, 0.01]
Skewness	-0.11	[-0.28, 0.07]
Kurtosis	-0.05	[-0.12, 0.02]

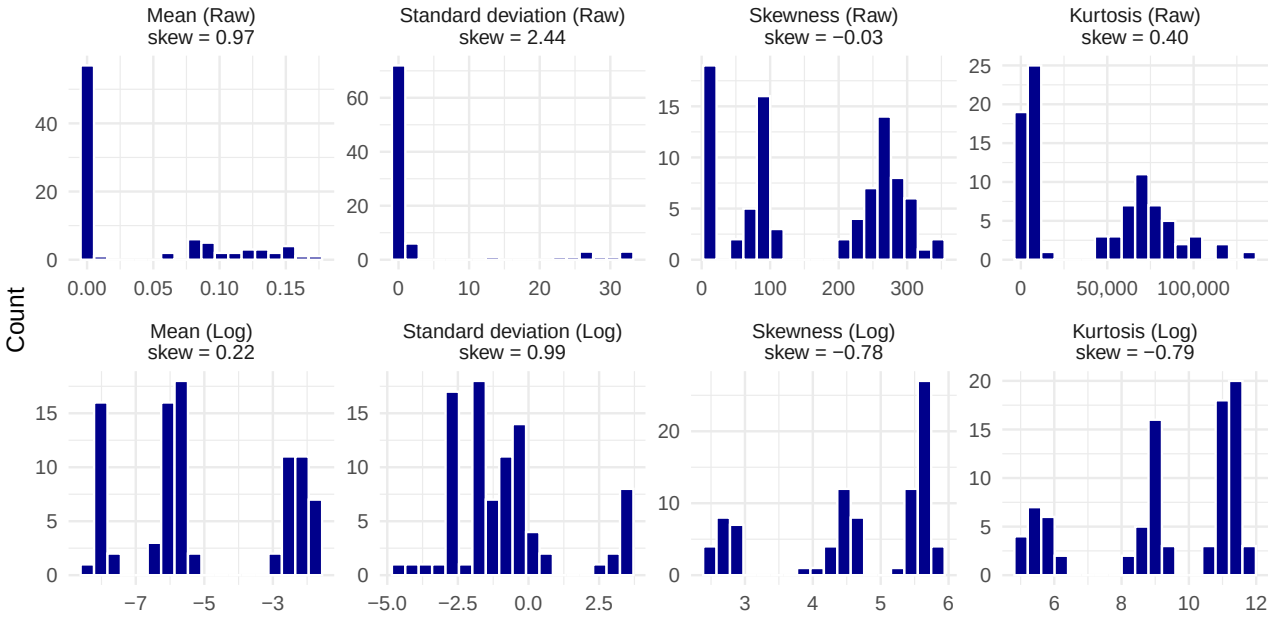


Figure 3: Distribution of each response on the raw scale (top) and log scale (bottom), with the sample skewness of each.

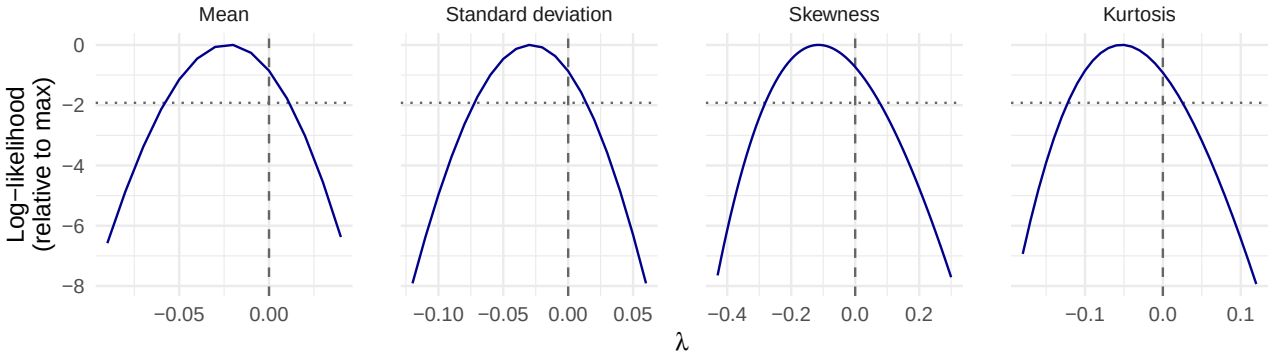


Figure 4: Box-Cox profile log-likelihood for each response under the cubic comparison model, relative to its maximum. Values of λ above the dotted line form the 95% interval; the dashed line marks $\lambda = 0$ (the log).